

WEEK 1 ASSIGNMENT

Introduction to Artificial Intelligence & Python

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TASK-1: INTRODUCTION TO ARTIFICIAL INTELLIGENCE

1.1 What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence.

AI allows computers to:

- Learn from data
- Recognize patterns
- Understand languages
- Make decisions
- Solve problems

Examples of AI:

- ChatGPT
- Google Assistant
- Face Recognition
- Recommendation Systems

1.2 Types of Artificial Intelligence

1. Artificial Narrow Intelligence (ANI)

ANI is AI designed to perform a specific task.

Examples:

- Voice assistants
- Face recognition
- Netflix recommendations

ANI cannot perform tasks outside its programmed purpose.

Image Space:

[Paste ANI Example Image Here]

2. Artificial General Intelligence (AGI)

AGI is a future type of AI that can perform any intellectual task like humans.

It can:

- Learn new skills
- Understand situations
- Solve different problems

Currently, AGI does not exist.



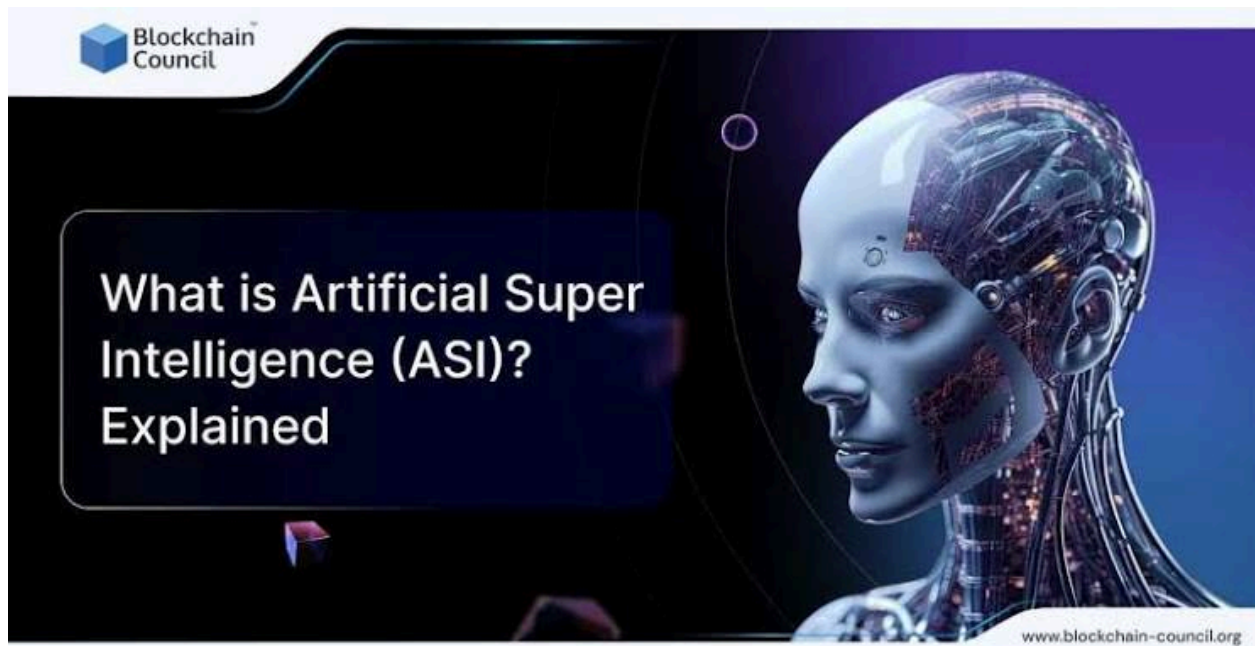
3. Artificial Super Intelligence (ASI)

ASI refers to AI that becomes more intelligent than humans.

It is only a theoretical concept.

Possible abilities:

- Advanced problem solving
- Scientific discoveries
- Complex decision making



1.3 AI vs Machine Learning vs Deep Learning

Artificial Intelligence (AI)

AI is the complete field of creating intelligent machines.

Example:

Self-driving cars.

Machine Learning (ML)

Machine Learning allows computers to learn from data and improve automatically.

Example:

Spam email detection.

Deep Learning (DL)

Deep Learning uses neural networks to process large amounts of data.

Example:

Face recognition.

AI	Machine Learning	Deep Learning
Broad concept	Part of AI	Part of ML
Creates intelligent systems	Learns from data	Uses neural networks
Example: Chatbots	Prediction systems	Image recognition

1.4 Real-Life Applications of AI

Healthcare

AI helps doctors in:

- Disease prediction
- Medical image analysis
- Patient monitoring

Education

AI provides:

- Personalized learning
- Online tutors
- Automated evaluation

Banking

AI is used for:

- Fraud detection
- Customer support
- Financial analysis

Entertainment

AI recommends content based on user interests.

Examples:

- Netflix
- YouTube
- Spotify

Transportation

AI helps in:

- Self-driving vehicles
 - Traffic management
 - Route planning
-

1.5 Advantages of AI

1. **Automation**
AI reduces human effort by performing repetitive tasks.
 2. **Fast Processing**
AI can analyze large amounts of data quickly.
 3. **Accuracy**
AI provides accurate results in many applications.
 4. **24/7 Availability**
AI systems can work continuously.
-

1.6 Limitations of AI

1. **High Cost**
Developing AI systems requires expensive resources.
2. **Job Replacement**
Automation may replace some human jobs.

3. **No Emotions**
AI cannot think or feel like humans.
 4. **Data Dependency**
AI requires large amounts of quality data.
-

TASK-2: SETTING UP AI DEVELOPMENT ENVIRONMENT

2.1 Introduction

Artificial Intelligence development requires a proper programming environment. Python is widely used in AI because it provides many useful libraries for data analysis, machine learning, and automation.

The required tools installed are:

- Python
 - VS Code
 - Jupyter Notebook
 - NumPy
 - Pandas
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2.2 Installing Python

Python is a popular programming language used for Artificial Intelligence and Machine Learning.

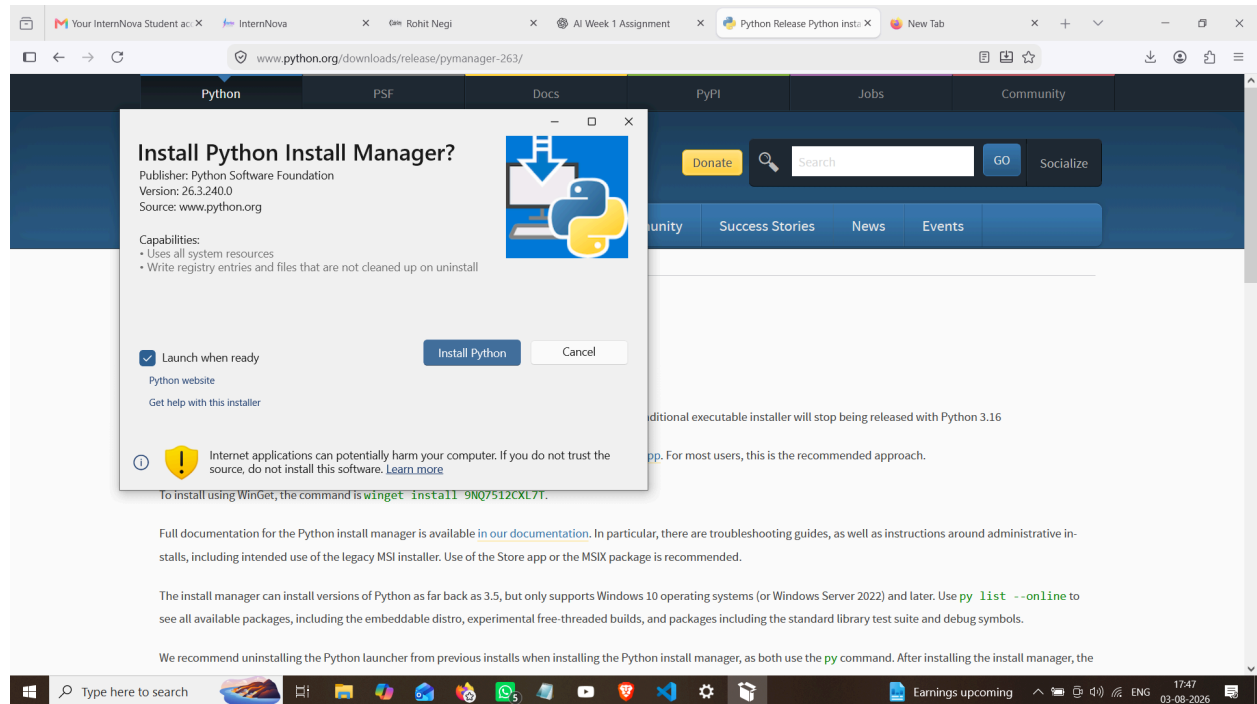
Installation Steps:

1. Download Python from the official website.
2. Install Python on the system.
3. Enable **"Add Python to PATH"** during installation.
4. Verify installation using:

`python --version`

Python helps developers write AI programs easily using different libraries.

Python Installation:



2.3 Installing VS Code

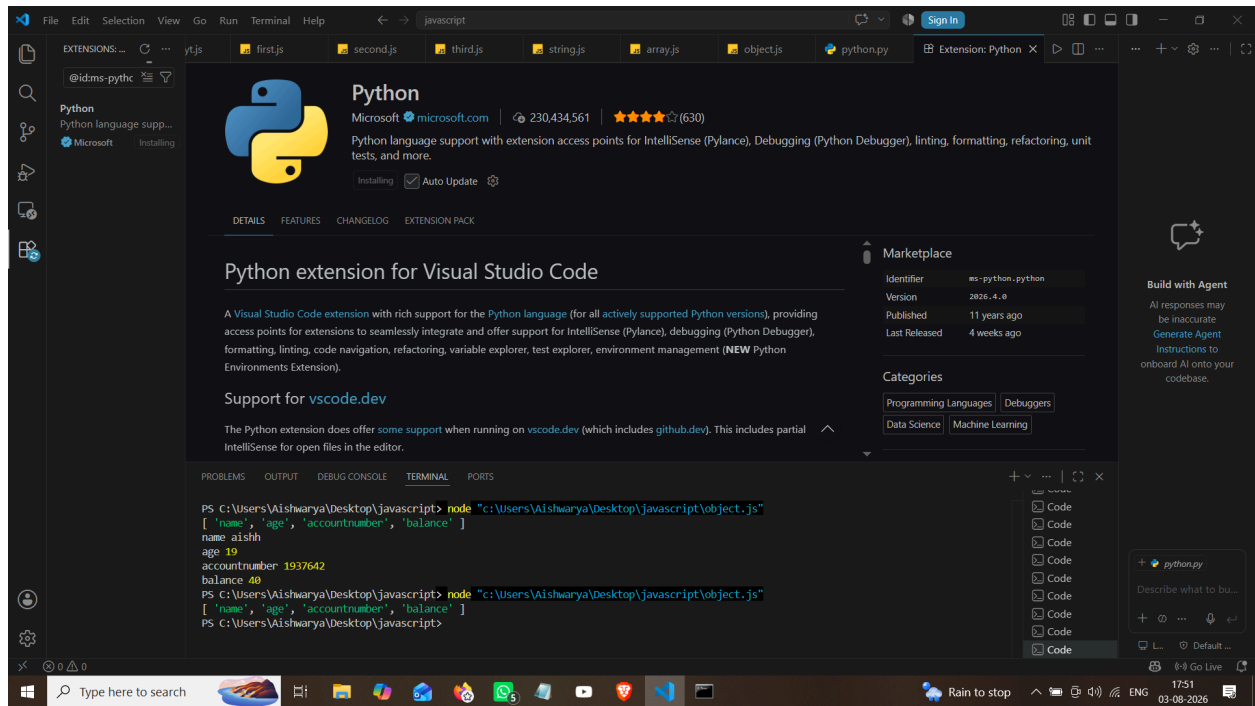
Visual Studio Code is a code editor used for writing and executing Python programs.

Features:

- Easy code writing
- Extensions support
- Debugging
- Integrated terminal

Python extension is installed in VS Code to run Python programs.

 **VS Code Interface with Python Extension**



2.4 Installing Jupyter Notebook

Jupyter Notebook is an interactive environment used for Python programming and AI experiments.

Installation command:

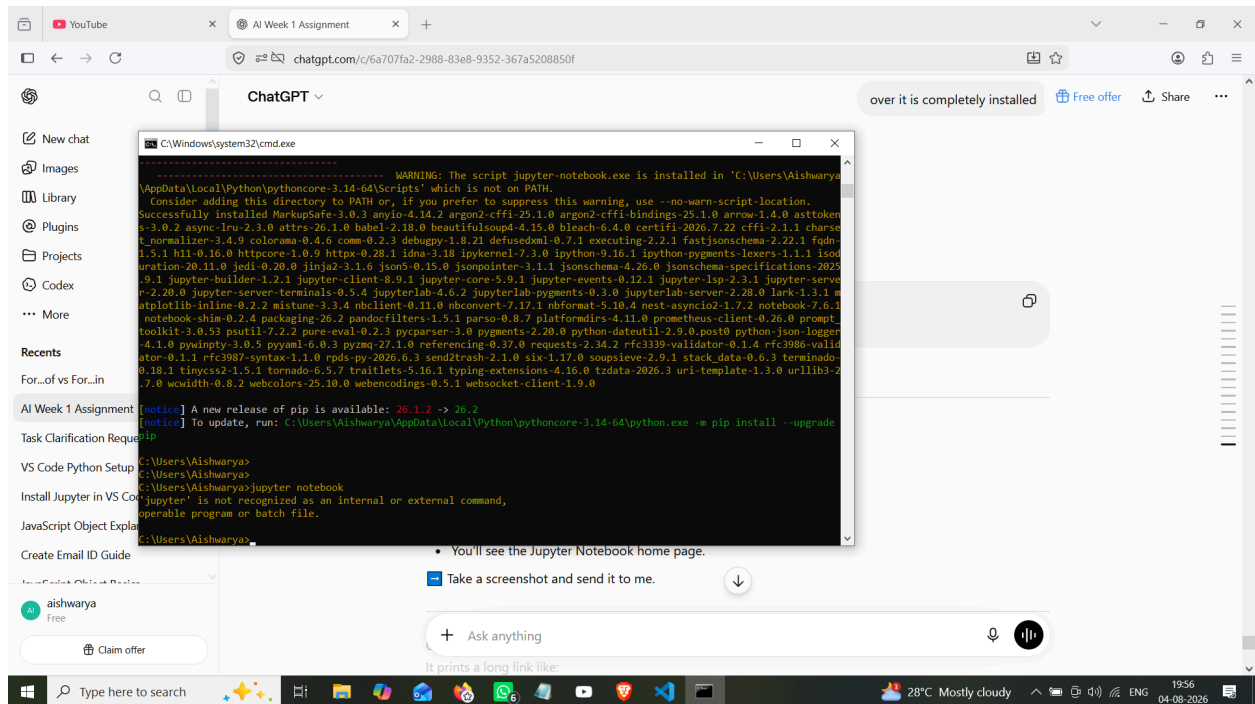
```
pip install notebook
```

To open Jupyter Notebook:

```
jupyter notebook
```

It helps users write code, execute programs, and view results easily.

 Jupyter installation



2.5 Installing AI Libraries

NumPy

NumPy is a Python library used for numerical calculations and handling arrays.

Installation:

```
pip install numpy
```

Pandas

Pandas is used for data analysis and data management.

2.6 Simple Python Program

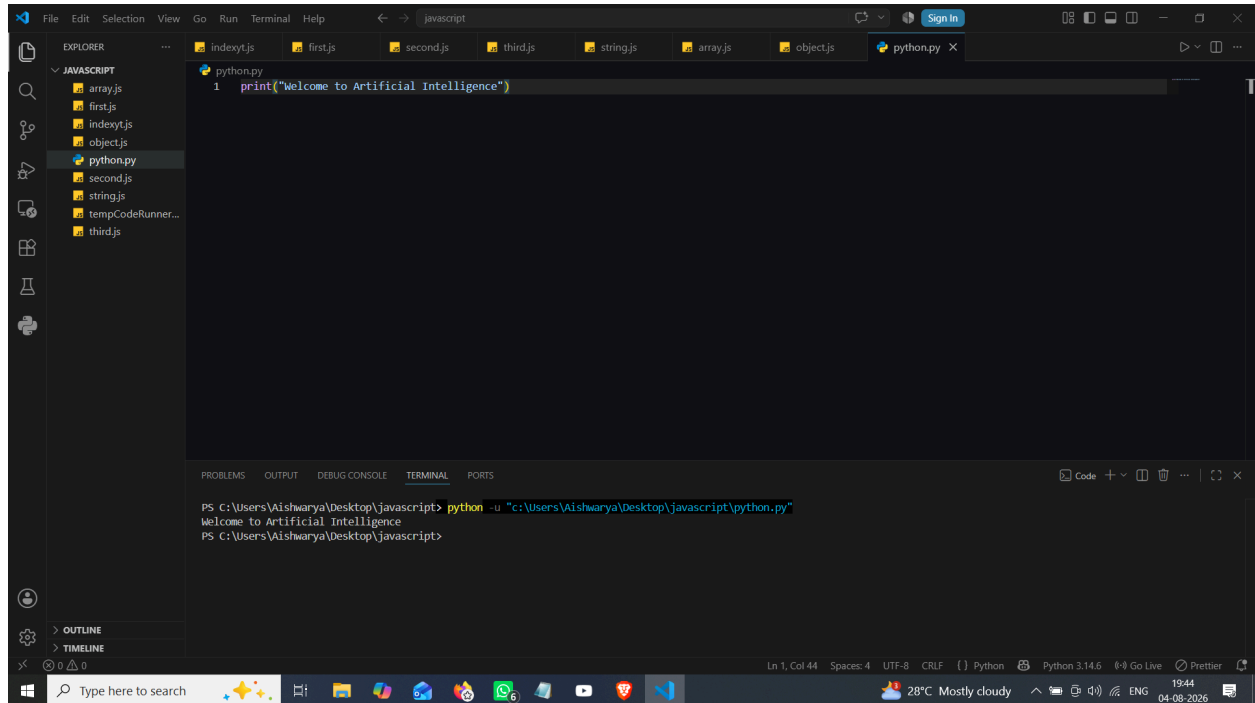
Program:

```
print("Welcome to Artificial Intelligence")
```

Output:

Welcome to Artificial Intelligence

This program displays a simple AI welcome message using Python.



The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named 'JAVASCRIPT' containing several files: 'array.js', 'first.js', 'index.js', 'object.js', 'python.py', 'second.js', 'string.js', 'tempCodeRunner...', and 'third.js'. The 'python.py' file is selected and open in the editor. The code in the editor is:

```
1 print("Welcome to Artificial Intelligence")
```

Below the editor, the terminal shows the command to run the script and its output:

```
PS C:\Users\Aishwarya\Desktop\javascript> python -u "C:\Users\Aishwarya\Desktop\javascript\python.py"
Welcome to Artificial Intelligence
PS C:\Users\Aishwarya\Desktop\javascript>
```

The status bar at the bottom indicates the file is at 'Ln 1, Col 44', uses 'Spaces: 4', 'UTF-8' encoding, and 'CRLF' line endings. It also shows the Python version as 'Python 3.14.6' and the editor theme as 'Prettier'.

TASK-3: PYTHON FOR AI

3.1 Introduction

Python is one of the most popular programming languages used in Artificial Intelligence. It is simple to learn and provides many libraries for AI and Machine Learning.

In this task, basic Python concepts are implemented:

- Variables

- Data Types
- User Input
- Operators

An AI-related example is created to predict whether a student passes or fails based on marks.

3.2 Variables in Python

A variable is a container used to store data values.

Example:

```
student_name = "Aishwarya"
```

```
marks = 85
```

Here:

- `student_name` stores the student's name.
- `marks` stores the student's marks.

Variables help AI programs store and process information.

3.3 Data Types in Python

Python supports different types of data:

1. Integer

Stores whole numbers.

Example:

```
marks = 90
```

2. Float

Stores decimal values.

Example:

```
percentage = 87.5
```

3. String

Stores text values.

Example:

```
name = "AI Student"
```

4. Boolean

Stores True or False values.

Example:

```
result = True
```

3.4 User Input

Python allows users to enter values using the `input()` function.

Example:

```
name = input("Enter your name: ")
```

User input helps AI systems collect information and make decisions.

3.5 Operators in Python

Operators are used to perform operations on values.

Arithmetic Operators:

Used for calculations.

Examples:

+

-

*

/

Comparison Operators:

Used for comparing values.

Examples:

>

<

==

Logical Operators:

Used to combine conditions.

Examples:

and

or

not

3.6 AI Example: Student Pass/Fail Prediction

This simple program predicts whether a student passes or fails based on marks.

Python Program:

```
name = input("Enter student name: ")
```

```
marks = int(input("Enter marks: "))
```

```
if marks >= 40:
```

```
    print(name, "has passed")
```

```
else:
```

```
    print(name, "has failed")
```

Explanation:

- The program takes student name and marks as input.
 - It checks the marks using an if-else condition.
 - If marks are 40 or above, it predicts "Pass".
 - Otherwise, it predicts "Fail".
-

3.7 AI Example: Temperature Conversion

AI systems used in weather forecasting need temperature data.

Example program:

```
celsius = float(input("Enter temperature in Celsius: "))
```

```
fahrenheit = (celsius * 9/5) + 32
```

```
print("Temperature in Fahrenheit:", fahrenheit)
```

This program converts Celsius temperature into Fahrenheit.

TASK-4: AI CASE STUDY – CHATGPT

4.1 Introduction

Artificial Intelligence is used in many real-world applications. One of the popular AI applications is **ChatGPT**, an AI-based chatbot that can understand and generate human-like text.

ChatGPT is developed using advanced AI techniques such as Natural Language Processing (NLP) and Deep Learning.

It can help users with:

- Answering questions
 - Writing content
 - Learning new concepts
 - Programming assistance
 - Problem solving
-

4.2 How AI is Used in ChatGPT

ChatGPT uses Artificial Intelligence to understand human language and provide meaningful responses.

The AI model is trained using a large amount of text data. It learns patterns, language structure, and relationships between words.

When a user enters a question, ChatGPT analyzes the input and generates a suitable response.

AI techniques used:

- Machine Learning
 - Deep Learning
 - Natural Language Processing (NLP)
 - Neural Networks
-

4.3 Input in ChatGPT

Input is the information provided by the user.

Examples:

- Questions
- Commands
- Text messages
- Requests for explanations

Example:

User Input:

"Explain Artificial Intelligence."

The AI system receives this text and processes its meaning.

4.4 Processing

Processing is the step where AI analyzes the user's input.

Steps involved:

1. Understanding the user's language.
2. Identifying important words and context.
3. Predicting the best possible answer.
4. Generating a response.

Deep learning models help ChatGPT understand complex language patterns.

4.5 Output

The final response generated by ChatGPT is the output.

Examples:

- Text answers
- Explanations
- Code suggestions
- Summaries

The output depends on the user's input and the information learned by the AI model.

4.6 Benefits of ChatGPT

1. Quick Information

ChatGPT provides fast answers to user questions.

2. Learning Support

Students can use it to understand difficult topics.

3. Programming Assistance

It helps developers by explaining code and solving programming problems.

4. Content Creation

It can help create articles, reports, and ideas.

5. 24/7 Availability

Users can access AI assistance anytime.

4.7 Limitations of ChatGPT

1. Incorrect Information

Sometimes AI may generate incorrect answers.

2. Requires Internet/Technology

Many AI services require digital access.

3. No Human Emotions

AI cannot truly understand feelings like humans.

4. Data Privacy Concerns

Users should avoid sharing sensitive information.

4.8 Working Flow of ChatGPT

User Input

|

↓

AI Model Processing

|

↓

Language Understanding

|

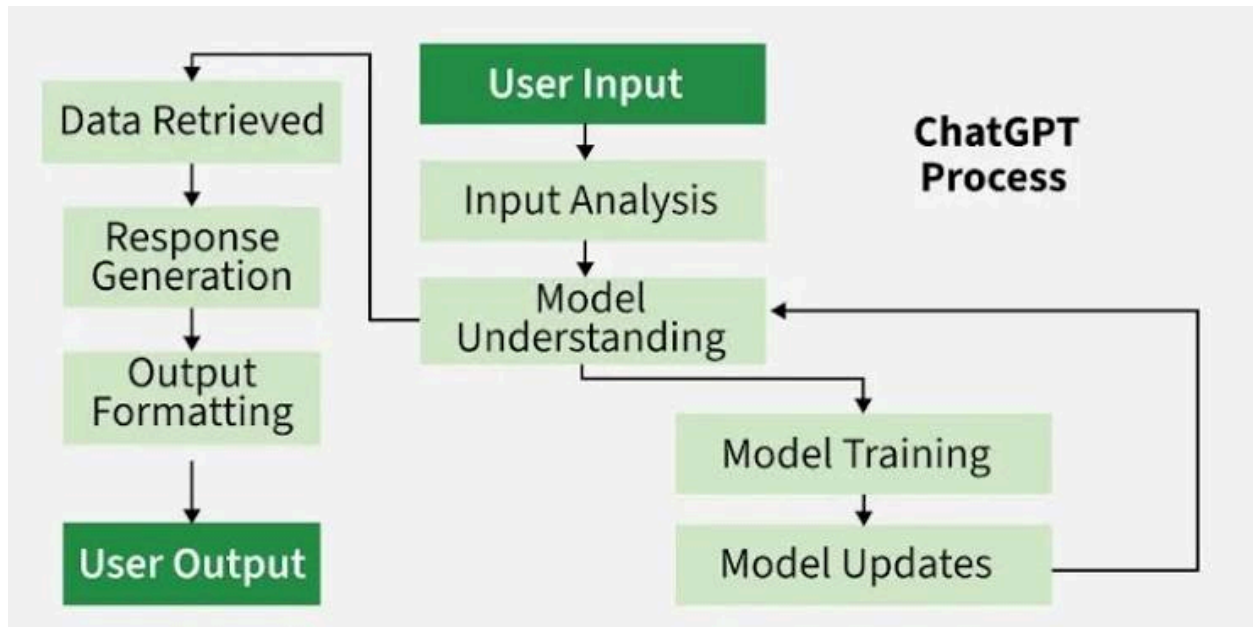
↓

Response Generation

|

↓

Output to User



TASK-5: MINI AI PROJECT – RULE-BASED AI CHATBOT IN PYTHON

5.1 Introduction

A chatbot is an AI program that communicates with users through text. A rule-based chatbot works using predefined rules and conditions.

In this project, a simple chatbot is created using Python **if-elif-else statements**. The chatbot responds according to the user's input.

This project does not use Machine Learning. It follows fixed rules created by the programmer.

5.2 Objective of the Project

The main objectives are:

- Understand basic AI concepts.
- Learn how chatbots work.

- Apply Python conditional statements.
 - Create a simple interactive AI program.
-

5.3 Working of Rule-Based Chatbot

The chatbot works in the following steps:

1. User enters a message.
2. The program checks the message.
3. It compares the input with predefined rules.
4. The chatbot gives a suitable response.

Example:

User: hello

Bot: Hello! Welcome.

5.4 Python Program

```
print("AI Chatbot Started")
print("Type 'bye' to exit")
```

```
while True:
```

```
    user = input("You: ").lower()
```

```
    if user == "hello":
        print("Bot: Hello! Welcome.")
```

```
    elif user == "how are you":
        print("Bot: I'm fine. How can I help you?")
```

```
    elif user == "what is ai":
        print("Bot: AI is the ability of machines to perform human-like tasks.")
```

```
    elif user == "bye":
        print("Bot: Goodbye!")
        break
```

```
else:  
    print("Bot: Sorry, I don't understand.")
```

5.5 Program Explanation

1. Input

The user enters a message through the keyboard.

Example:

You: hello

2. Processing

The chatbot checks the user's message using:

- if statement
- elif statement
- else statement

It matches the input with stored rules.

3. Output

The chatbot provides a suitable reply.

Example:

Bot: Hello! Welcome.

5.6 Sample Output

AI Chatbot Started

You: hello

Bot: Hello! Welcome.

You: how are you

Bot: I'm fine. How can I help you?

You: what is ai

Bot: AI is the ability of machines to perform human-like tasks.

You: bye

Bot: Goodbye!

5.7 Advantages of Rule-Based Chatbot

- Easy to create and understand.
 - Provides quick responses.
 - Useful for simple customer support.
 - Helps beginners understand AI logic.
-

5.8 Limitations of Rule-Based Chatbot

- Cannot learn from new conversations.
 - Works only with predefined rules.
 - Cannot understand complex questions.
-

5.9 Future Improvements

The chatbot can be improved by adding:

- Machine Learning algorithms
- Natural Language Processing
- Speech recognition
- Large language models

FINAL REPORT CONCLUSION

Artificial Intelligence is changing the way technology works in different fields. Through this assignment, we learned AI fundamentals, Python environment setup, basic Python programming, AI applications, and chatbot development.

Python provides a simple and powerful platform for beginners to start learning Artificial Intelligence and Machine Learning.